Team	NCM Innovators (Minnette Johnson, Carolann Mora, Peter Olayiwola)
Driver	D1 – Improve staff inventory search, filtering, rentability and location across four stores
Package version	v1.2
Submitted	

1. Purpose

D1 extends the staff Fleet interface across four Golden Bike Shop locations. It covers lookup, reservations, returns and safety checks, workshop release, transfers, overdue rentals, e-bike eligibility, and idle stock. To balance advance bookings (FR-03/04) with uncertain return readiness (FR-18/19), the team approved a waitlist and optional free upgrade. A projected return alone cannot confirm a booking. Owner approval is pending.

2. The Flows Your Driver Touches

Flow inventory source: v1.0 requirements and Week 2 owner replies A1–A15.

Flow	Actor	New, changed or at risk	What the driver does to it
UC-01 Search and verify immediate rentability	Counter clerk / Rental Agent	Changed from v1.0	Type/size lookup; distinguish location from “can rent now”; accurate local count (FR-03, FR-16).
UC-02 Check requested period and reserve bicycle	Rental Agent	Changed from v1.0	Check conflicts and projected returns before saving booking (FR-03, FR-04, FR-18, FR-19).
UC-03 Record return, inspect and release bicycle	Rental Agent; Mechanic / check actor TBD	Changed from v1.0	Prevent immediate re-rental before wipe-down/safety check and approved service release (FR-07, FR-18).
UC-04 Record workshop intake and mechanic release	Mechanic; rental desk observes	Changed from v1.0	Workshop blockers visible in Fleet; mechanic alone marks ready (FR-07).
UC-05 Dispatch and receive inter-store transfer	Sending clerk; receiving clerk	New in v2.0 proposal	Show On Van, destination, rough arrival; exclude in-transit stock (FR-17).
UC-06 Identify late rentals and contact renter	Rental Agent	New in v2.0 proposal	Show customer and due-back; support manual call once one hour late (FR-19).
UC-07 Check e-bike telemetry eligibility	Rental Agent	Unchanged but at risk	Apply v1.0 battery/fault/15-minute freshness guard to lookup (FR-06, NFR-03).
UC-08 Compare cross-store fleet location and counts	Store Clerk / Manager	Changed from v1.0	Display consistent branch, status and eligible local count (FR-16, FR-17, NFR-01).
UC-09 Review idle bicycles for management action	Store Manager	New, definition pending	Identify bikes idle for weeks; criteria require owner validation (FR-20).

Owner answers received after Week 2

Owner post and date	What he said	What it changes

No verified additional replies after the Week 2 submission (Nov. 1).	Most recent verified owner posts: Professor Morgan Worrell, Oct. 28 (A1–A8) and Oct. 31 (A9–A15).	No new post-Week-2 requirements are asserted.

3. Use-Case Diagram

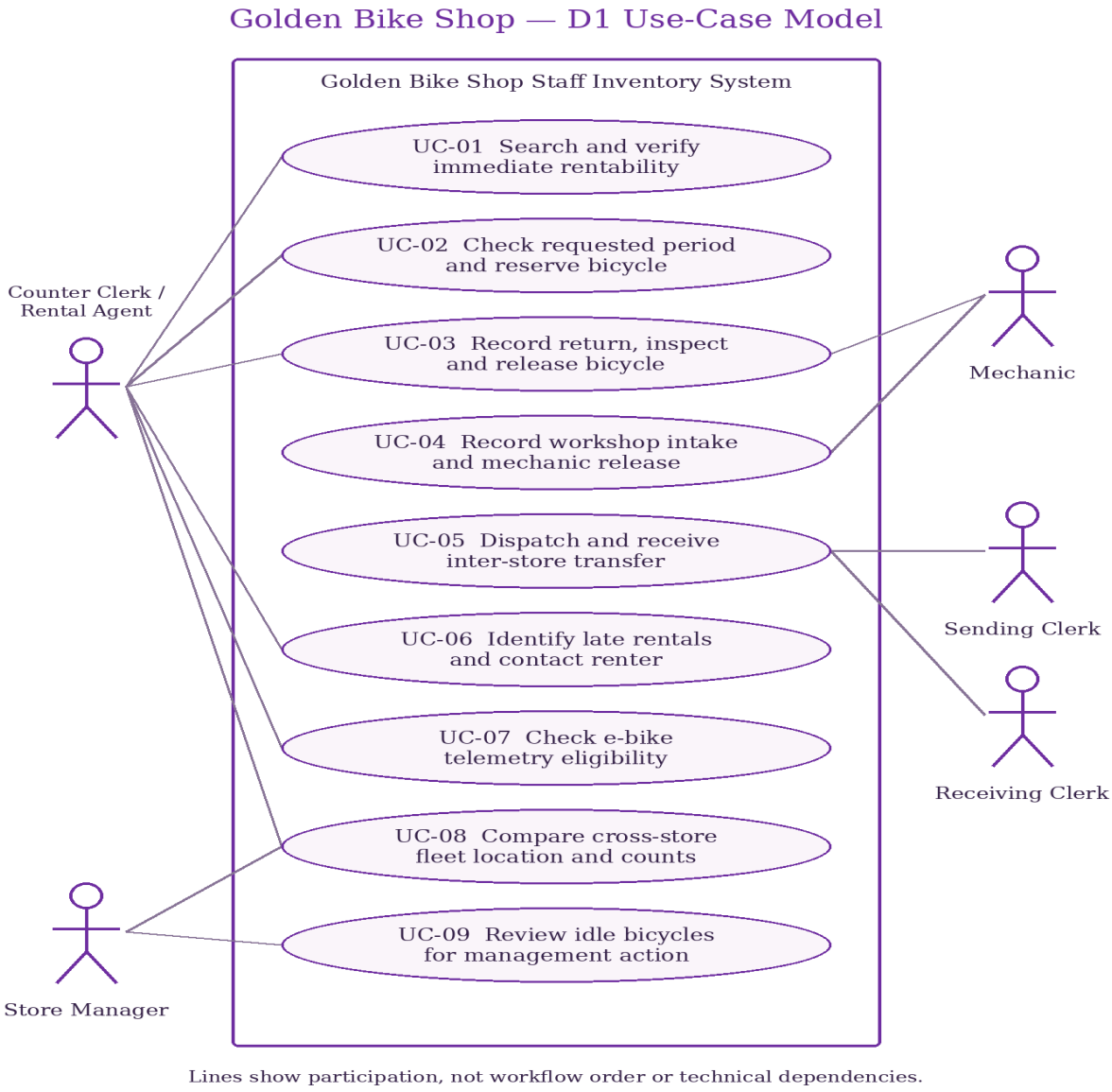


Figure 1. The nine driver flows and participating staff roles within the Golden Bike Shop Staff Inventory System; customer requests are mediated by staff in this diagram.

4. Use Cases

UC-01, UC-02, and UC-03 are the most critical because they verify availability, confirm reservations, and determine when returned bikes can be rented again.

UC-01: Search and Verify Immediate Rentability — MINNETTE (fully dressed)

Primary actor	Counter Clerk / Rental Agent.
Stakeholders and what each wants	Counter staff: find accurate local eligible stock quickly. Rental agents: avoid promising an unavailable bicycle. Customers: receive reliable information. Mechanics: keep service bikes out of stock. Professor Morgan Worrell (owner role): reduce preventable lost rentals and verification phone calls.

Preconditions	Staff is authorized to use the Fleet view (baseline FR-12); selected branch exists; central inventory can be queried. A query may return zero or indeterminate availability; no assumption is made that every source is current.
Trigger	A customer asks whether a bicycle of a particular type and size can leave the selected branch now.
Main path	1. Open Fleet and select the physical branch the customer asks about. [FR-16; Worrell A3, A14, Oct. 28/31] 2. Search for bikes matching both type and frame size. [FR-16; Worrell A14, Oct. 31] 3. Show matches at their verified current location; flag transit separately from home branch. [FR-16, FR-17; Worrell A3, A5, A11, Oct. 28/31] 4. Check active rental/reservation, workshop/retired, transit, return-check and e-bike blockers. [FR-03, FR-06, FR-07, FR-17, FR-18; Worrell A1, A4, A9–A11, Oct. 28/31] 5. Count confirmed rentable-now matches at the selected branch; show reasons for blocked local stock. [FR-16; Worrell A1, A3, A14, Oct. 28/31] 6. Review individual status and applicable e-bike battery, fault and freshness. [FR-06, FR-16; Worrell A3, Oct. 28] 7. Communicate verified current eligibility; recheck future requests in UC-02. [FR-03, FR-04; Worrell A1, A6, Oct. 28]
Extensions	1a. Require a branch before showing its count. [FR-16] 2a. Show zero matches and zero rentable now; no size substitution without staff input. [FR-16] 3a. Show On Van destination and ETA; count at neither branch until receipt. Dispatch/receipt roles remain open. [FR-17; Worrell A5, A11] 3b. Mark conflicting or missing location/eligibility evidence unverified. [NFR-01; proposed safety handling pending validation] 4a. Show overdue, workshop, retired or unchecked bikes as blocked, with reason. [FR-03, FR-07, FR-18, FR-19; Worrell A1, A9, A10, A12] 4b. Exclude e-bikes below 30% battery, with a blocking fault, or with readings older than 15 minutes. [FR-06, NFR-03; baseline v1.0] 5a. Exclude unverified eligibility; prompt for a check. Exact feedback awaits validation. [NFR-01; proposed exception] 7a. Route future requests to UC-02; due-back alone does not guarantee readiness. [FR-03, FR-04, FR-18; Worrell A6, A10]
Guaranteed at the end	A branch-scoped search result is presented, including a confirmed-rentable count or a clearly unverified/no-results outcome; no bicycle is reserved, released from service, or marked available merely by viewing the screen. The existing business and e-bike blockers remain enforced.
Owner sources	Professor Morgan Worrell, Week 2 forum replies: Oct. 28, A1/A3/A4/A5/A6; Oct. 31, A9/A10/A11/A14. Baseline v1.0 FR-03/06/07, NFR-01/03 and §5.4.
Written by	Minnette Johnson — UC-01 contributor.

UC-02: Check Period Availability and Create Reservation — PETER (fully dressed)

Primary actor	Rental Agent. The customer requests the booking through the agent; the gateway supports payment but does not initiate this staff use case.
Stakeholders and what each wants	Rental staff want usable future-booking information; customers want a reservation the shop can honour; mechanics retain release authority; the owner wants fewer lost rentals without false promises. [A1, A4, A6, A9–A12]
Preconditions	An authorized Rental Agent and a valid pickup branch are identified; customer details and a start/end period can be supplied. Central reservation, workshop, return-check and telemetry evidence can be checked. Authorization is a baseline requirement, not an implemented MVP capability. Future readiness policy remains open. [FR-03/04/12; Conflict 1]
Trigger	A customer asks the Rental Agent to reserve one or more bicycles for a specified pickup branch and time window.
Main path	1. Enter customer, pickup branch, start/end, type and size. Reject end ≤ start. [FR-03/04/16; v1.0 §5.3; A14, Oct. 31] 2. Check period overlaps and workshop blocks. Separate projected due-backs from verified eligibility; projections alone cannot confirm. [FR-03/04/18/19; A6, Oct. 28; A10/A12, Oct. 31; Conflict 1 pending] 3. Select eligible bikes and review branch, period and per-bike rates; selection does not confirm. [FR-04; v1.0 §§2, 5.3] 4. Recheck authoritative return, service, transit and e-bike eligibility. Atomically save the reservation and bike items with overlap protection. Payment pending is not a forecast hold. [FR-03/04/06/07/17/18; NFR-02/03; v1.0 §5.3] 5. Request deposit via gateway; store its reference, not card details. [FR-13; NFR-08; v1.0 §§3.3, 5.3] 6. Confirm once after deposit succeeds and the reservation is saved, preserving per-bike period constraints;

	see 5a/6a for failure. [FR-04/13; NFR-02; v1.0 §5.3] 7. After commit, invalidate affected cache entries and release the confirmation event. [NFR-01; v1.0 §§4, 5.3] 8. Return the committed reference, bikes, branch and period. Staff communicate it; pickup checks still apply. [FR-04/06/07/13; A4/A9/A10; v1.0 §3.3]
Extensions	1a. Explain missing customer/branch or invalid period; return to step 1 without confirmation. [FR-04] 2a. Label forecast-only bikes projected; offer confirmed alternatives or defer until Week 4 policy. No tentative booking state. [A6/A10/A12; Conflict 1] 2b. Show ineligibility, overlap, overdue, unchecked or transit blocker; return to selection without promising readiness. [FR-03/07/17/18/19] 4a. If a concurrent booking wins, roll back this attempt and requery; no silent substitution or partial booking. [FR-04; NFR-02; all-or-none handling proposed for validation] 4b. Exclude e-bikes below 30%, with active fault, or telemetry older than 15 minutes. Offer eligible alternatives; unknown is unconfirmed. [FR-06; NFR-03] 5a. On declined/cancelled deposit, release the hold without confirmation. Reconcile uncertain gateway outcomes before charging again. [FR-13; v1.0 §§3.3, 5.3] 6a. If save fails or confirmation is uncertain, report no confirmation; reconcile gateway and database before retry/refund. No distributed atomicity. [FR-13; NFR-01/02; proposed recovery acceptance] 7a. On post-commit cache or delivery failure, keep the booking; revalidate against authoritative data and recover delivery without rebooking. [NFR-01/02; v1.0 §4]
Guaranteed at the end	Success: one confirmed central reservation with one item per selected bicycle, agreed rates, gateway reference and no overlapping active commitments. Otherwise no false confirmation; failed-attempt holds are released or explicitly unresolved for reconciliation. Pickup safety and mechanic-only release remain enforced. Proposed exception handling requires team validation.
Owner sources	Morgan Worrell, Week 2 thread: Oct. 28 A1/A4/A6 and Oct. 31 A9–A12/A14; v1.0 §§2–5 (FR-03/04/06/07/12/13; NFR-01/02/03/08). Week 2 C-01–C-06 / FR-16–19. Source thread: https://worldclass.regis.edu/d2l/le/391702/discussions/threads/2369570/View
Written by	Peter Olayiwola

UC-03: Record Return, Inspect, and Release Bicycle — CAROLANN (fully dressed)

Primary actor	*Rental Agent (records the return and the post-return check). Mechanic is a supporting actor only when the check finds a fault; the workshop release itself is UC-04.* *Open: the owner has not confirmed who performs the cleaning and quick safety check. This case assumes the Rental Agent until the Week 4 forum answers it.*
Stakeholders and what each wants	*Rental Agent: check a bike in quickly, never lose track of it, and know exactly when it may go out again.* *Returning customer: rental closed promptly, with an accurate return time and no damage blamed on them that was not found at check-in.* *Next customer (walk-in or future booking): receives a bike that is clean, safe and ready, not one that was merely "back."* *Mechanic: faults reach the workshop with enough detail to act on; nobody outside the workshop clears a workshop hold.* *Staff at the other three stores: a returned-but-unchecked bike never appears in their rentable counts.* *Professor Morgan Worrell (owner role): no dirty or unsafe bike is re-rented, and a checked bike returns to service quickly so it can earn again.*
Preconditions	*Staff member is signed in with the rental-desk role (baseline FR-12).* *The bike has an active or overdue rental record and a readable fleet ID.* *Current return-check status, workshop status and, for e-bikes, telemetry can be queried.*
Trigger	*A customer physically presents a rented bicycle at a store counter to end the rental.*
Main path	*1. Scan or enter fleet ID; retrieve the bike's current rental, including overdue rentals, never future bookings. [FR-03, FR-19]* *2. Close the rental; record step 1 time, actual receiving store and minutes late (overdue at ≥1 hour). Link any damage found in step 4; payment is out of scope. [FR-03, FR-19; Worrell A12, Oct. 31]* *3. Update current location, not home store. Set Returned – Awaiting Check centrally: physical stock only, projected in UC-02. After the passing check is committed, recheck period eligibility before confirming (Conflict 1). [FR-16, FR-18; Worrell A10, Oct. 31]*

	4. Clean and inspect using a fixed checklist that includes cleaning. Record pass/fail per item; any failure triggers a workshop hold (4a). [FR-18, FR-07; Worrell A9, A10, Oct. 31] *5. Show e-bike battery, faults and telemetry time for information; they do not change the check outcome. Refresh telemetry at step 7. [FR-06, NFR-03; baseline v1.0]* *6. The person who checked the bike confirms under their own sign-in. Require all items to pass; record actor and time. [FR-18, FR-12]* *7. Recheck workshop, fresh FR-06 telemetry and current overlaps. If clear, mark Rentable at receiving store; otherwise show Checked – Not Rentable and retest when the blocker clears. [FR-03, FR-06, FR-07, FR-18]* *8. Commit centrally, clear cache and reflect status in UC-01/02 across stores. Until commit is verified, count as rentable nowhere (8a). [FR-16, NFR-01]*
Extensions	*1a. If no active rental matches the ID, close nothing and search by customer. If unmatched, mark Returned – Unmatched and hold. [NFR-01; proposed handling]* *1b. For lost/stolen bikes, skip normal return, flag for manager follow-up and mark Missing, excluded from all counts. [Proposed; owner to confirm in Week 4]* *2a. Record lateness and close any UC-06 follow-up already opened at one hour overdue. [FR-19; Worrell A12, Oct. 31]* *2b. For a different receiving store, update current location only; use UC-05 for any transfer back. [FR-16, FR-17; Worrell A5, A11]* *4a. Record damage/fault, apply workshop hold and route to UC-04. Only a mechanic clears it. [FR-07; Worrell A4, Oct. 28; A9, Oct. 31]* *4b. If the check cannot finish, retain Returned – Awaiting Check and list it as pending. Time alone never releases it. [FR-18; Worrell A10]* *4c. Without a mechanic, retain the hold; rental staff cannot override it. [FR-07]* *5a. Battery below 30%, blocking fault or telemetry older than 15 minutes blocks rental even if the check passes. [FR-06, NFR-03; baseline v1.0]* *6a. Reject check recording without the rental-desk role; keep the bike held. [FR-12]* *6b. Reject non-mechanic workshop release; use UC-04. [FR-07; Worrell A4]* *7a. If pickup starts before checks finish, alert staff and keep readiness unconfirmed (Conflict 1, UC-02). [FR-03, FR-04; Worrell A6, A10]* *8a. If the central update is unverified, count as rentable nowhere and show "update pending" at the receiving store. [NFR-01; proposed fail-safe]*
Guaranteed at the end	*Success: the rental is closed with its return time, receiving store and lateness recorded; the bicycle is located at the receiving store; it is Rentable only after a recorded, passing check by authorized staff with no workshop hold or e-bike blocker; every store sees the same status.* *Minimal (if any step fails): the bicycle stays not rentable. A return check or workshop hold is never cleared implicitly, by elapsed time, or by viewing a screen.*
Owner sources	*Professor Morgan Worrell, Week 2 forum replies: Oct. 28, A4/A5/A6; Oct. 31, A9/A10/A11/A12. Baseline v1.0 FR-03/06/07/12, NFR-01/03 and 5.5.* *Disclosure: the actor who performs cleaning and the quick safety check is unresolved; this case assumes the Rental Agent pending the Week 4 forum.*
Written by	*Carolann Mora — UC-03 contributor. See Section 5 for rewrites.*

Brief use cases

Use case	Actor	What happens, in one short paragraph	Requirement IDs
UC-04 Workshop intake and mechanic release	Mechanic; rental desk	*A mechanic records workshop intake after a fault, inspection or scheduled maintenance. The In Service hold excludes the bike at every store and shows the reason and expected completion. Only a mechanic clears the hold; then apply UC-03 step 7. Linking the paper job sheet remains open for Week 4.*	FR-07; A4/A9
UC-05 Inter-store bicycle transfer	Sending clerk; receiving clerk	*The sending clerk records dispatch. Show On Van, destination and estimated arrival; count the bike at neither store. On receipt, update its location and count it only after eligibility checks. Dispatch/receipt ownership and alert timing remain open.*	FR-17; A5/A11

UC-06 Follow up overdue rental	Rental Agent	The Rental Agent reviews bicycles out today, their renters and due-back times. The system marks an unreturned bicycle late at or after due-back plus 60 minutes; the agent initiates the existing phone call. Expected return supports planning, not automatic readiness. No automated SMS is added. [A6, Oct. 28; A12/A14, Oct. 31]	FR-19; A6/A12/A14
UC-07 Verify e-bike telemetry eligibility	Rental Agent	Show battery, faults and telemetry age. Exclude e-bikes below 30% battery, with blocking faults, or with telemetry older than 15 minutes (FR-06).	FR-06, NFR-03; v1.0
UC-08 Review cross-store Fleet	Store Clerk / Manager	Compare location and status across four stores; distinguish transit from local stock and count only eligible bikes at the selected branch.	FR-16, FR-17, NFR-01; A3/A5/A14
UC-09 Investigate idle fleet	Store Manager	*List idle bikes by location, type, size and days idle for manager review. The system makes no transfer or price change. Idle threshold and rental/retail stock scope await owner confirmation.*	FR-20; A2/A13

5. Ask for Every Reading

Individual AI prompts and output excerpts are in the Week 3 Records.

Use case and step	Reading the tool gave	Verdict	Step before and after	Team member
UC-01 step 1	A. Employee's home branch; B. Customer's requested branch.	A: Wrong but plausible; B: Intended	Before: "Open Fleet at the store." After: Select the physical branch whose stock the customer asks about.	Minnette
UC-01 step 2	A. Type OR size; B. Type AND size.	A: Wrong but plausible; B: Intended	Before: "Filter by type and size." After: Select both type and frame size as required matching criteria.	Minnette
UC-01 step 3	A. Home branch; B. Verified current location, including transit.	A: Wrong but plausible; B: Intended	Before: "Show bikes at this store." After: Display verified current location and separate On Van status.	Minnette
UC-01 step 4	A. Physically present; B. No active blockers; C. Badge color alone.	A: Wrong but plausible; B: Intended; C: Not plausible	Before: "Check whether each bike is available." After: Test rental, service, transfer, return check and e-bike blockers.	Minnette
UC-01 step 5	A. All stores; B. Confirmed matches at selected branch.	A: Wrong but plausible; B: Intended	Before: "Show the available count." After: Count confirmed rentable NOW, scoped to selected branch.	Minnette
UC-01 step 6	A. Any reading today; B. Battery, fault and 15-minute freshness.	A: Wrong but plausible; B: Intended	Before: "Review e-bike battery." After: Display battery/fault/freshness; apply v1.0 FR-06 threshold (4b).	Minnette
UC-01 step 7	A. Lookup books the bike; B. Booking is separate (UC-02).	A: Wrong but plausible; B: Intended	Before: "Tell customer they can book." After: Communicate confirmed status; run UC-02 to request booking.	Minnette
UC-02 steps 1–8 (Peter)	Per step: intended / wrong but plausible. 1. Exact branch and dates / home branch or duration only. 2. Verified period eligibility / due-back guarantees readiness. 3. Select bikes and rates / selection confirms. 4. Fresh check and protected write / cached or partial booking. 5. Gateway reference / store card details. 6. Confirm after deposit and save / payment initiation suffices. 7. Notify after commit / notify before commit. 8. Retain pickup checks / confirmation overrides safety. Not plausible: font or color authorizes booking.	Intended / wrong but plausible / not plausible, as labelled.	Before, steps 1–8: "Enter booking details"; "Show available bikes"; "Select bikes and rates"; "Save the booking"; "Take the deposit"; "Confirm the booking"; "Update inventory and notify"; "Tell the customer." After, steps 1–8: enter exact branch and dates; separate forecasts from verified eligibility; selection does not confirm; recheck and save with overlap protection; retain gateway reference only; confirm after successful deposit and save; notify after commit; communicate the recorded booking and retain pickup checks. See §4 for full steps.	*Peter*
UC-03 step 1	*A. Open rental for scanned ID.* *B. Overdue rental not found.* *C. Future reservation retrieved.* *D. Visual match, no ID entry.* *E. Customer's rentals, not bike's.*	*A: Intended; B, C: Wrong but plausible; D, E: Not plausible*	*Before: "…identifies the returned bicycle by its fleet ID … retrieves that bicycle's active rental."* *After: Retrieve the current rental by fleet ID, including overdue rentals; exclude future bookings.*	*Carolann*
UC-03 step 2	*A. Closes at identification.* *B. Damage can't attach after close.* *C. Return time = time recorded.*	*A, C: Intended; B, D, E, F: Wrong but plausible*	*Before: "…closes the rental and records the return time, the receiving store and whether the return was late…"* *After: Record the time at identification and actual receiving store; overdue at ≥1 hour. Link damage; exclude payment.*	*Carolann*

	D. Late = any minute past due. *E. Store = agent's login store.* *F. Includes billing/payment.*			
UC-03 step 3	*A. Only current location changes.* *B. Home store reassigned.* *C. In physical stock, not rentable.* *D. Hidden from every count.* *E. Still reservable for the future.* *F. Other stores update later.*	*A, C: Intended; B, D, E, F: Wrong but plausible.*	*Before: "…sets the bicycle's current location to the receiving store and its status to Returned – Awaiting Check…"* *After: Update current location only. Count physical stock; show only projected in UC-02 while unchecked.*	*Carolann*
UC-03 step 4	*A. Agent must do it personally.* *B. Anyone works; agent records.* *C. Cleaning is a check item.* *D. Cleaning not recorded.* *E. Fixed checklist.* *F. Free-text note.* *G. Fail creates workshop holds.* *H. Fail only noted.*	*B, C, E, G: Intended; A, D, F, H: Wrong but plausible*	*Before: "The Rental Agent cleans the bicycle, performs the quick safety check and records each check item…"* *After: Use a fixed checklist including cleaning; record pass/fail per item and hold on failure.*	*Carolann*
UC-03 step 5	*A. Information only.* *B. The agent must acknowledge.* *C. Stale reading reused at step 7.* *D. Empty fields for non-e-bikes*	*A: Intended; C: Wrong but plausible; B, D: Not plausible*	*Before: "If the bicycle is an e-bike, the system shows its battery level, any blocking fault and … last telemetry reading."* *After: Show telemetry for information; refresh it at step 7.*	*Carolann*
UC-03 step 6	*A. "All passed" overrides a fail.* *B. Blocked unless all pass.* *C. "Who" = whoever is logged in.* *D. Another agent confirms later.* *E. "When" = confirmation time*	*B, E: Intended; A, C, D: Wrong but plausible*	*Before: "The Rental Agent confirms that every check item passed, and the system records who … and when."* *After: The person who checked confirms under their own sign-in; require all items to pass.*	*Carolann*
UC-03 step 7	*A. Any future booking blocks.* *B. Only overlapping-now blocks.* *C. Blocked bike stays Awaiting Check.* *D. Separate blocked status.* *E. Applied at home store.* *F. Charging = unchecked.* *G. Re-test runs once only.*	*B, D: Intended; A, C, E, F, G: Wrong but plausible*	*Before: "The system re-tests rentability … If none applies, it sets the status to Rentable at the receiving store."* *After: Check current overlaps and blockers; mark Checked – Not Rentable if blocked and retest when clear.*	*Carolann*
UC-03 step 8	*A. Waits for all four stores.* *B. Effective at central update.* *C. One store down blocks all.* *D. Not even projected in UC-02.* *E. Real-time push to screens.*	*B: Intended; A, C, D, E: Wrong but plausible*	*Before: "…makes the new status visible at all four stores, and only then includes the bicycle in UC-01 lookups and UC-02…"* *After: Commit centrally, then clear cache and update UC-01/02 across stores.*	*Carolann*

Steps rewritten	Minnette: 7 draft steps clarified (1–7). Peter: Rewrote 8 UC-02 steps. Carolann: 8 of 8 UC-03 steps rewritten. Of 46 readings, 15 intended, 27 wrong but plausible, 4 not plausible.
Kind of step that drew the most wrong readings	Lookup: location, current rentability and telemetry freshness. Booking: requested period, payment and commit order. Returns: status changes (steps 3, 7, 8) drew 13 of 27 wrong readings, mainly about when and where a status takes effect. Actor and signoff steps (4, 6) followed.

6. Conflicts and How Each Was Resolved

Conflict 1

Requirements in conflict	FR-03/FR-04 permit requested-period search/reservation; FR-18/FR-19 require return-check and late-return awareness; FR-16 requires truthful rentable-now counts. NFR-02 forbids overlapping commitments. A bike due at 2 p.m. may not be ready for a 3 p.m. booking. [A6/A10/A12]
Stakeholders on each side	Rental agents and customers value advance booking; mechanics and waiting customers need safe, actually ready bicycles. The owner bears both lost-sales and broken-promise costs. These are competing interests, not evidence that named teammates took opposing positions.
Options considered	1. Do nothing: staff manually reconcile uncertainty; low change cost, existing failures persist. 2. Confirm from due-back forecasts: more bookings, risk of late/unchecked bikes. 3. Wait for completed return checks before all future confirmations: reliable current readiness, unnecessarily restricts advance trade. 4. Separate projected planning from confirmed eligibility: retain useful forecasts without portraying them as guaranteed stock; extra workflow/testing cost.
Approach	Negotiated resolution (SWEBOK ch. 1 §3.4). Minnette proposed separating projected returns from confirmed availability on November 3.
Trade made, and why	Keep projections visible for planning and allow eligible advance bookings. A forecast alone does not confirm readiness; extra verification protects customer trust and NFR-02. Peter's proposal — Automated waitlist and optional free upgrade (team approved; owner approval pending): For forecast-only requests, send a waitlist acknowledgment by email/SMS without promising inspection clearance, confirming, or taking a deposit. Confirm through UC-02 only after verifying a suitable bike. If the original bike is late or fails inspection, offer an eligible upgrade at the original price with customer consent, if available; otherwise notify the customer and offer another time or branch.
Who agreed	Minnette Johnson and Carolann Mora approved Peter's proposal in team Slack on November 7, 2026. Week 4 owner question: For a 2 p.m. due-back and 3 p.m. pickup, do you approve the waitlist, email/SMS acknowledgment and optional free upgrade? When should staff offer an alternative, and who authorizes an upgrade? Owner approval is pending before changing UC-02 or its requirements.

Conflict 2

Requirements in conflict	*Side A: FR-16 (an accurate count of bikes rentable now at the selected branch) and NFR-01 (consistent values across stores).* *Side B: FR-17 (show a bike On Van with its destination and rough arrival so staff can plan and tell customers).* *The conflict: a bike on the van is physically at neither store, so any count that includes it at one store is wrong for some period, while leaving it out of both makes counts drop during every transfer.*
Stakeholders on each side	*Side A: counter clerks and store managers, who need a number they can promise a customer from; the owner, who wants fewer verification calls between stores (Worrell A3, A14).* *Side B: sending and receiving clerks, who need a transfer workflow that adds as few steps as possible at a busy counter; customers, who want to know a bike is coming instead of being told "none."* *Both sides share one interest: customers should not be sent to a store for a bike that is not there.*
Options considered	*1) Do nothing: v1.0 keeps the bike at the sending store until someone edits it. Phantom stock at the sender; staff keep phoning. Rejected.* *(2) Count it at the sending store until receipt: no undercount, but the sender promises a bike that has left. Rejected; this is the false promise D1 exists to remove.* *(3) Count it at the destination from dispatch: the destination promises a bike that has not arrived and may be delayed. Rejected for the same reason.* *(4) Count it at neither store; show On Van, destination and rough arrival separately: counts stay true; cost is a temporary undercount and a required receipt step.* *(5) Option 4 for "rentable now," but let future-period checks at the destination see the bike after its expected arrival as projected, not confirmed (ties to Conflict 1).*
Approach	*Both approaches from SWEBOK §3.4. For the rentable-now count we propose a negotiated resolution (option 4) to be agreed with the owner. For future bookings we let the requirement vary by type of request (option 5): a walk-in question gets only confirmed stock, while a future-period check may see a projected arrival, labelled as projected.* *Dispatch and receipt are the authoritative events: the bike leaves the sender's count at dispatch and joins the destination only at recorded receipt. Every store should see each event within the same propagation window as any other status change (NFR-01).*

Trade made, and why	*Count only bikes confirmed rentable at the store. Show in-transit bikes with their destination and estimated arrival.* *Gave up: transfer bikes count nowhere while in transit, so branch counts dip during transfers, and the receiving clerk must record each arrival.* *Why: A missed rental costs less than promising a bike that has not arrived.*
Who agreed	*Team: agreed by Carolann Mora and Peter Olayiwola at the Week 3 huddle (Nov. 4). Minnette was absent; her written confirmation is pending.* *Owner: NOT YET AGREED. Question for the Week 4 forum: "While a bike is on the van between stores, may staff count it as available at either store? Who should record dispatch and receipt, and how long past the expected arrival should staff be alerted?"*

7. What the 601 Architecture Must Do

UC-01–03 are assessed against the MSSE 601 modular core, PostgreSQL, Redis, telemetry and transaction rules.

Use case	What the architecture must do	Step where it would struggle, and why
UC-01 (Minnette)	Present cross-store Bike/location/status plus rental, repair, transfer, return-check and e-bike telemetry eligibility as one trustworthy staff answer. Use v1.0 PostgreSQL/telemetry snapshots, enforce 15-minute rule, and preserve 95% search <=3s (NFR-05).	Steps 3–5: v1.0 location/status lists do not model On Van or post-return cleaning distinctly; Redis may briefly show a pre-transfer or pre-return result. A correct near-real-time count needs consistent meaning and cache freshness, not simply a faster screen.
UC-02 (Peter)	UC-02 steps 2/4/6 require central requested-period checks, fresh service/return/transit evidence, per-bike overlap protection and one committed result. Preserve the final 601 Spring Boot modular core/BFF, PostgreSQL ReservationItem/RateCard, Redis read cache, outbox and separate telemetry snapshot. Apply FR-06 at query time, not merely a periodic stale flag; preserve 95% search ≤3 s / submissions ≤5 s. Kafka is the final baseline's telemetry broker, superseding earlier RabbitMQ wording.	Step 2: Period/location history exists; period search, readiness policy and completed-check evidence are missing. Step 4: Revalidate in PostgreSQL and enforce overlap constraints to avoid stale-cache races. Steps 5–7: The gateway is outside the database transaction; reconcile failures and notify after commit. Login/RBAC, workshop APIs and return/transit semantics remain gaps. No redesign or passed tests are claimed.
UC-03 (Carolann)	*Keep Returned – Awaiting Check, Workshop Hold and Rentable as distinct states in the PostgreSQL source of truth; allow the move to Rentable only when a recorded passing check exists and no workshop or FR-06 blocker applies.* *Enforce roles: rental desk records checks; only a mechanic clears a workshop hold.* *Record who and when for every state change; commit it in PostgreSQL first, then invalidate cached availability and release the status event through the outbox, so no store sees the bike as rentable before the commit*	*Steps 3 and 7: v1.0 5.5 moves a returned bike straight to Available. There is no state for "back but unchecked," so FR-18 has nowhere to live and the transaction rules would allow a rental in that gap. This is a gap in the state model, not a performance limit.* *Step 8: the Redis cache can serve a pre-return result at another store until it is refreshed, so correctness depends on cache invalidation, not only on the database write.* *Extension 4a: workshop intake starts on a paper job sheet, so a hold may be entered late. The architecture can only enforce a hold that has been recorded.*

8. Changes to the Package

Change log entries this week

Change	Version	Req IDs	What changed	Why (source)	Owner
W3-01 [DRAFT]	*v1.2 proposed*	*FR-03/04/18/19*	*FR-03/FR-04 clarified: projected returns are shown separately from confirmed rentable stock; no future reservation is confirmed against an unreturned or unchecked bike until the owner rules (Week 4). Team approved the waitlist and optional free upgrade proposal under Conflict 1. Owner approval is pending; the current confirmation rule remains in effect.*	*Conflict 1; UC-02 (Peter); Worrell A6/A10/A12*	*Carolann + Peter*

W3-02 [DRAFT]	*v1.2 proposed*	*FR-16/17; NFR-01*	*FR-17 amended: On Van is a status distinct from both branches. FR-16 amended: the rentable-now count excludes On Van bikes; dispatch and receipt are the events that move a bike between counts. Owner agreement pending. .*	*Conflict 2; UC-01 step 3; Worrell A5/A11*	*Carolann*
W3-03 [DRAFT]	*v1.2 proposed*	*FR-16; FR-06*	*FR-16 reworded to "bikes confirmed rentable now at the selected branch," matching both criteria; FR-06 language aligned with the battery/fault/15-minute rule used in UC-01 4b and UC-03 5a.*	*UC-01 step ambiguity review (Section 5)*	*Minnette / Carolann*
W3-04 [OPEN]	*v1.2 proposed*	*FR-07/18*	*FR-18 amended: a returned bike enters Returned – Awaiting Check and is not rentable until a recorded, passing check by authorized staff. FR-07 restated: only a mechanic clears a workshop hold. v1.0 §5.5 direct Returned → Available transition removed. The actor for the cleaning/safety check remains OPEN (Week 4).*	*Worrell A4/A9/A10; UC-03 pending*	*Carolann*
W3-05 [DRAFT]	*V1.2 proposed*	*FR-18/FR-19/FR-06*	*Adds Checked – Not Rentable for a bike that passed its check but has a blocker; defines return time and overdue (one hour); the check is confirmed only by the person who did it, with every item passed; step 7 uses a fresh telemetry reading.*	*Worrell A4/A9/A10; UC-03 pending*	*Carolann*

Week 3 audit findings

Finding	Status	How it was dealt with, or why it moved
A-01: Status consistency	*Partly closed; remainder moved to Week 4*	*Partly closed in Week 3: UC-01 and UC-03 now separate location, lifecycle state and blockers (W3-02, W3-04). Moved to Week 4: the workshop intake/release event, because it depends on the owner's answer about the paper job sheet.*
A-02: Exceptions	*Partly closed; remainder moved to Week 4*	*Partly closed in Week 3: the UC-01 and UC-03 extensions now cover late return, incomplete check, stale telemetry, unmatched returns and failed synchronization, each failing safe. Moved to Week 4: owner acceptance of the fail-safe hold and the lost-bike path (UC-03 1b).*
A-03: Narrow-screen usability	*Moved to Week 4 (previous plan)*	*Stays in Week 4 as planned: it needs real staff task observations and WCAG checks, which use cases cannot supply.*
A-04: Current location / transfer	*Moved to Week 5 (previous plan)*	*Stays in Week 5 as planned. Week 3 progress: On Van is now a defined state with a counting rule (Conflict 2, W3-02). Dispatch/receipt responsibility and timing go to the Week 4 forum first.*
Repository	*Week 3 (v1.2) package committed to msse-portfolio/golden-bike-shop: https://github.com/morac32/msse-portfolio/commit/64fd7a28ca121b70440a85ef9babbf8f399c2ae1*	

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